Single Proportion Inference (Solutions)

Inference on a binary population

We've seen how to use a binomial null distribution when studying p.

Powerplant example: Null is $Binom(300, 0.5)$ and p-value is found with `dbinom`.

Next, we'll discuss a **Z test** with a **N(0, 1)** null.

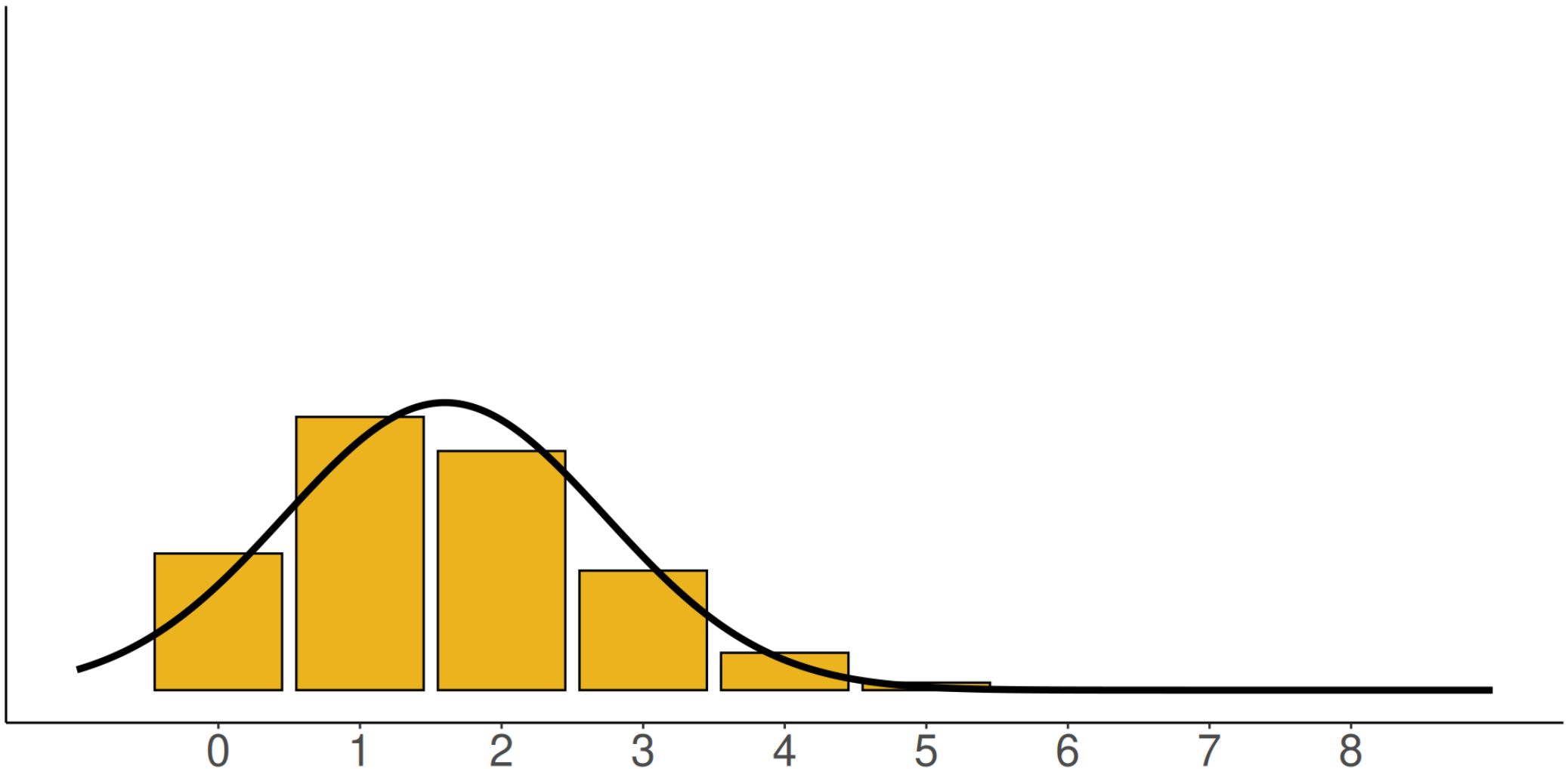


How can we use a normal curve as a model for p?

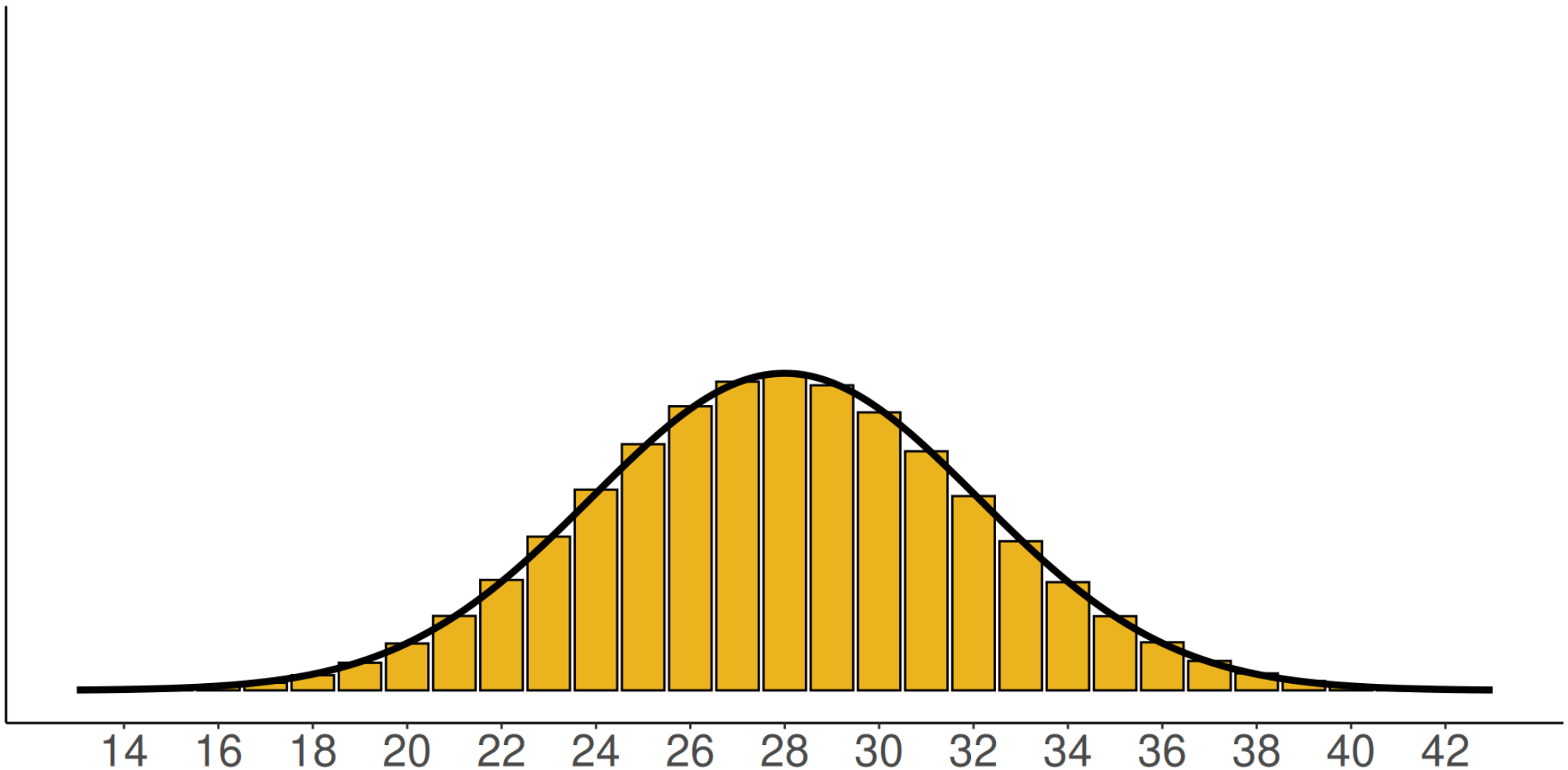
A binomial resembles a normal when n is large and p is close to 0.5.

$$X \dot{\sim} N\left(np, \sqrt{np(1-p)}\right)$$

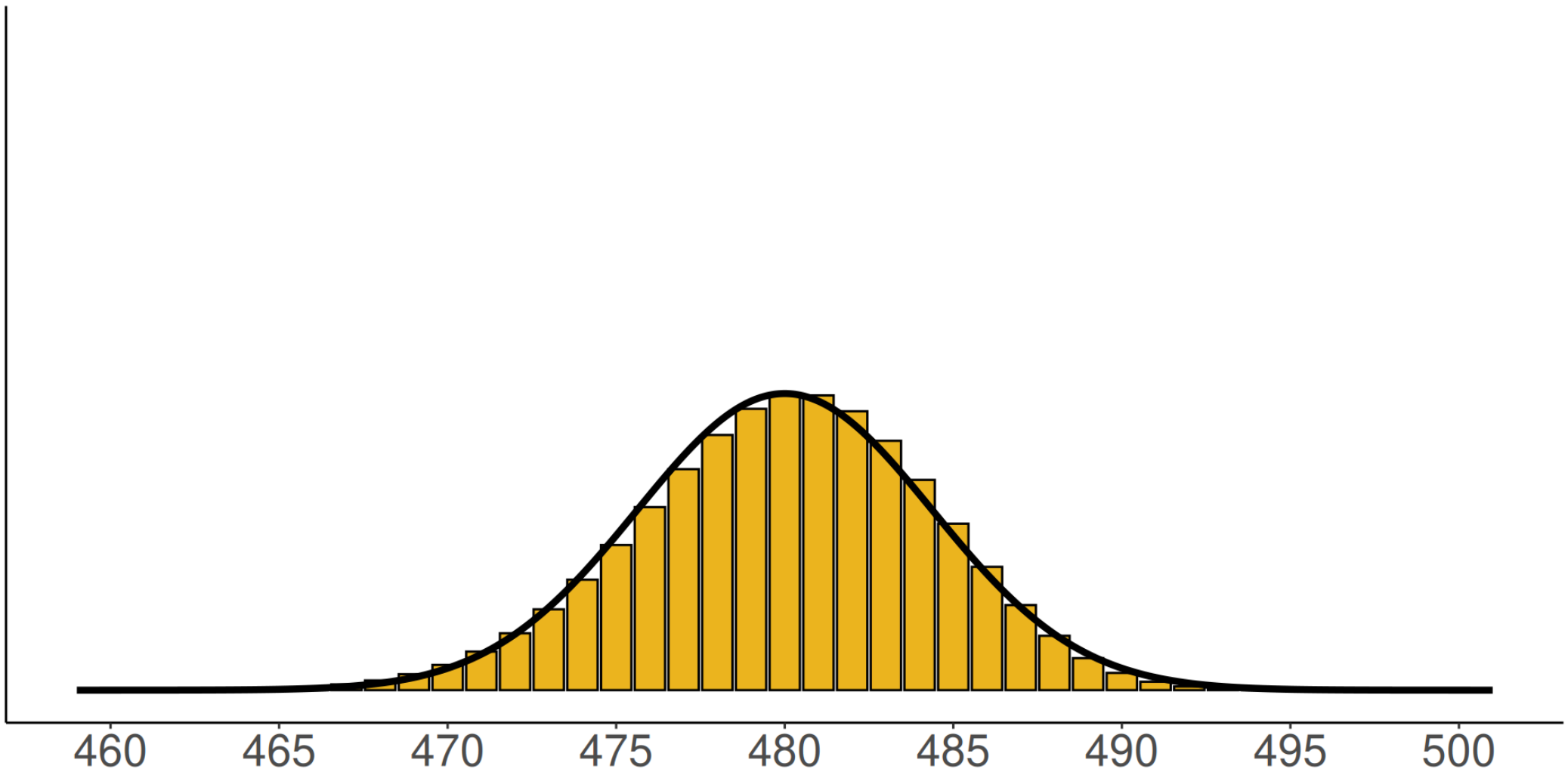
$Binom(8, 0.2)$ versus $N(1.6, 1.13)$



$Binom(70, 0.4)$ versus $N(28, 4.1)$



$Binom(500, 0.96)$ versus $N(480, 4.38)$



Why this normal curve? For any $X \sim Binom(n, p)$,

$$E(X) = np, \qquad sd(X) = \sqrt{np(1-p)}$$

Use these as the parameters to the normal curve.

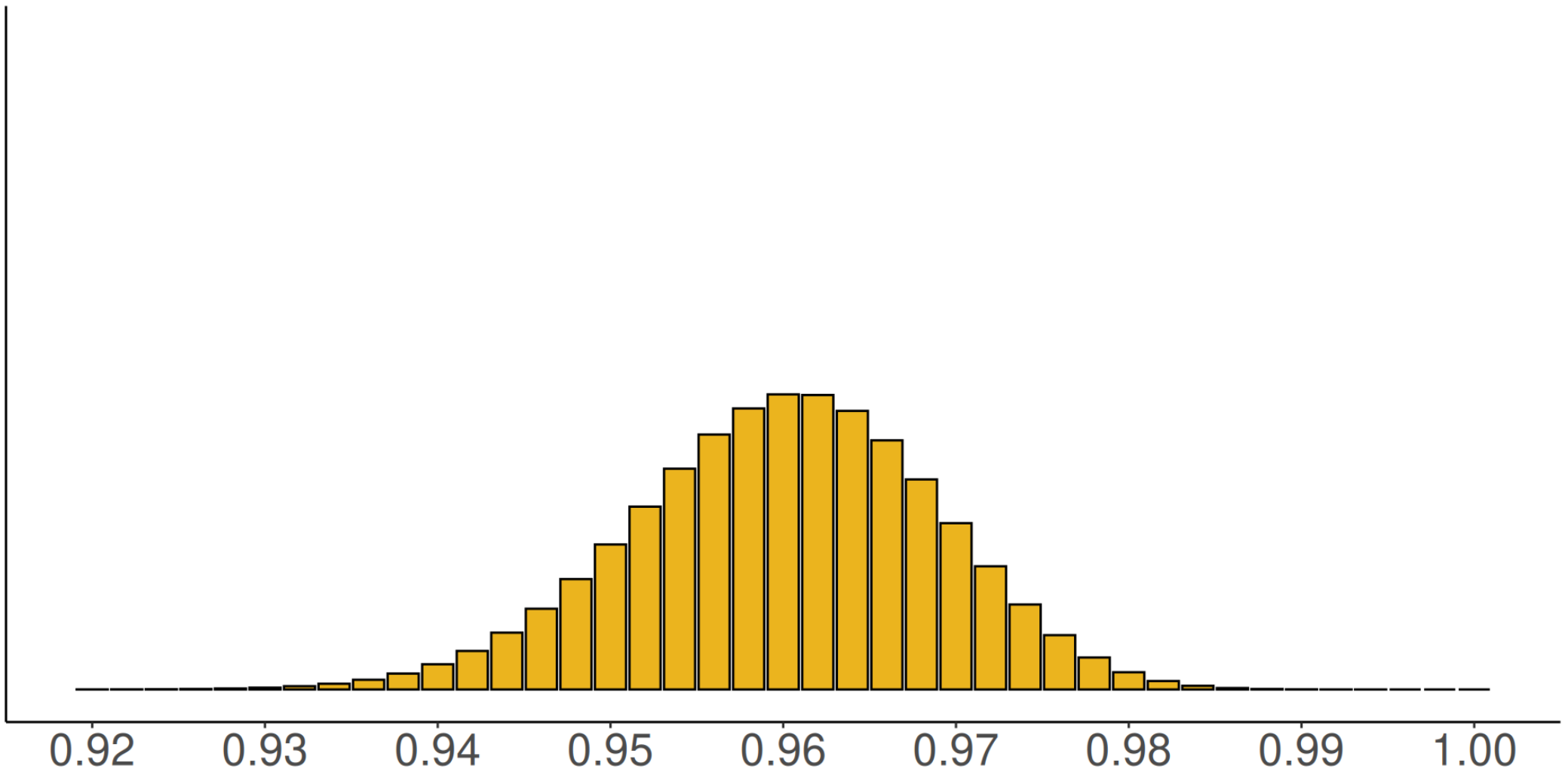
But check $np(1-p) \geq 10$!

X, the binomial, is the count of successes. Now let's consider $\hat{p}$, the sample proportion of successes.

$$\hat{p} = \frac{X}{n}$$

If X is binomial, $\hat{p}$ looks a lot like a binomial. It also looks like a normal curve for large n.

$$\hat{p} \;=\; Binom(500, 0.96)/500$$



If $X \sim Binom(n, p)$, this is true in general:

$$E(X) = np, \qquad sd(X) = \sqrt{np(1-p)}$$

$$E(\hat{p}) = p, \qquad sd(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

If $X \sim Binom(n, p)$ AND $np(1-p) \geq 10$, this is true:

$$X \dot{\sim} N(np, \sqrt{np(1-p)})$$

$$\hat{p} \dot{\sim} N\left(p, \sqrt{\frac{p(1-p)}{n}}\right)$$

So, we can use a normal curve for inference on p.

Use the $N(0, 1)$ for

- A null distribution (Z test)
- Quantiles for building a Z confidence interval

Exercises

- What is the variance of a binomial random variable? What is the variance of $\hat{p}$?
- What is the relationship between the two variance terms?
- What are the advantages / disadvantages to doing inference with simulation, a binomial null, or a $N(0, 1)$ null?



Solutions/Hints

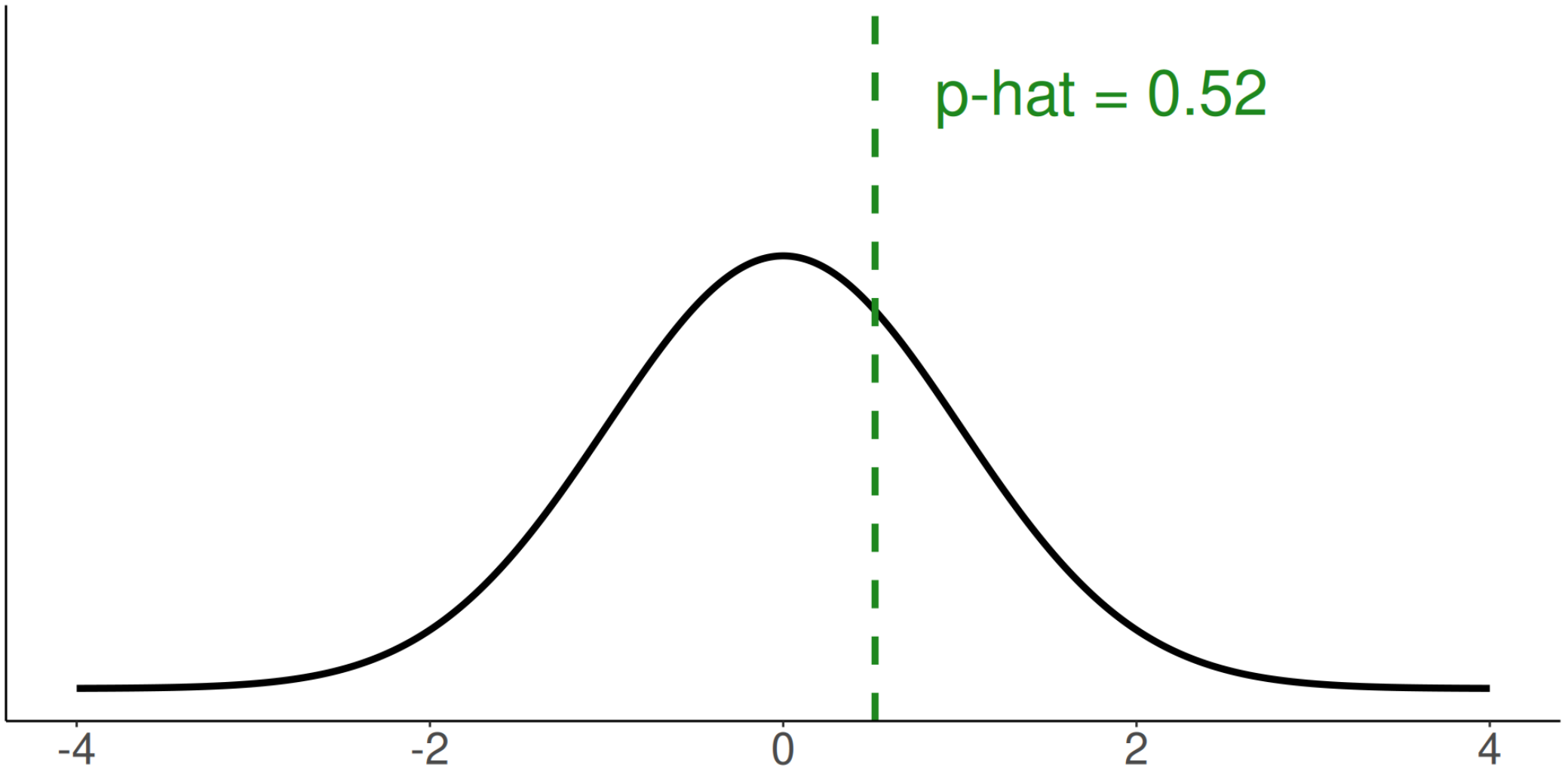
- The variance of a binomial is $np(1-p)$ and the variance of $\hat{p}$ is $\frac{np(1-p)}{n}$
- The variance terms differ by a factor of n^2.
- Simulation and binomial approaches require fewer assumptions than the normal Z test. However, the Z test allows us to use a standardized reference distribution for many different contexts.

Let's redo the powerplant births example with a Z test.

$$H_0 : p = 0.5 \quad \text{versus} \quad H_A : p \neq 0.5$$

Where does the $Z \sim N(0, 1)$ null come from?

Should we compare the observed 0.52 directly to the $N(0, 1)$ curve? No - we need to standardize.



$$\hat{p} \dot{\sim} N\left(p, \sqrt{\frac{p(1-p)}{n}}\right)$$

turns into

$$\hat{p} \dot{\sim} N\left(0.5, \sqrt{\frac{0.5(1-0.5)}{300}}\right)$$

With a little bit of standardization:

$$Z = \frac{\hat{p} - 0.5}{\sqrt{\frac{0.5(1-0.5)}{300}}} \dot{\sim} N(0,1)$$

The left side is our **test statistic**.

$$Z = \frac{\hat{p} - 0.5}{\sqrt{\frac{0.5(1-0.5)}{300}}} \approx \frac{\text{data - null}}{\text{error}}$$

- Numerator: compare observed $\hat{p}$ to null 0.5
- Denominator: error of $\hat{p}$ (also called **standard error**)

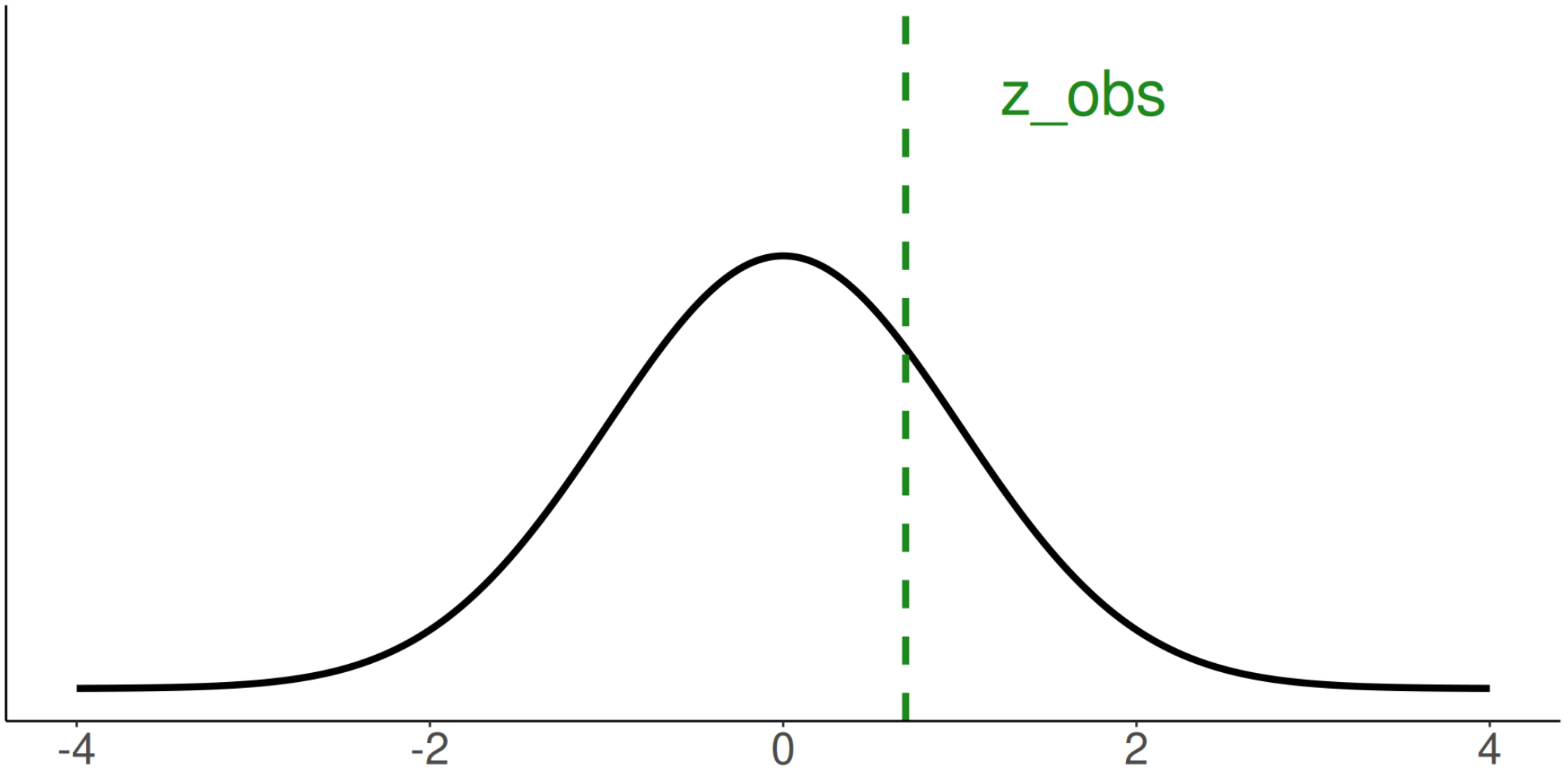
A test statistic comes from our data and has a *known distribution* under H_0.

Our observed test statistic is

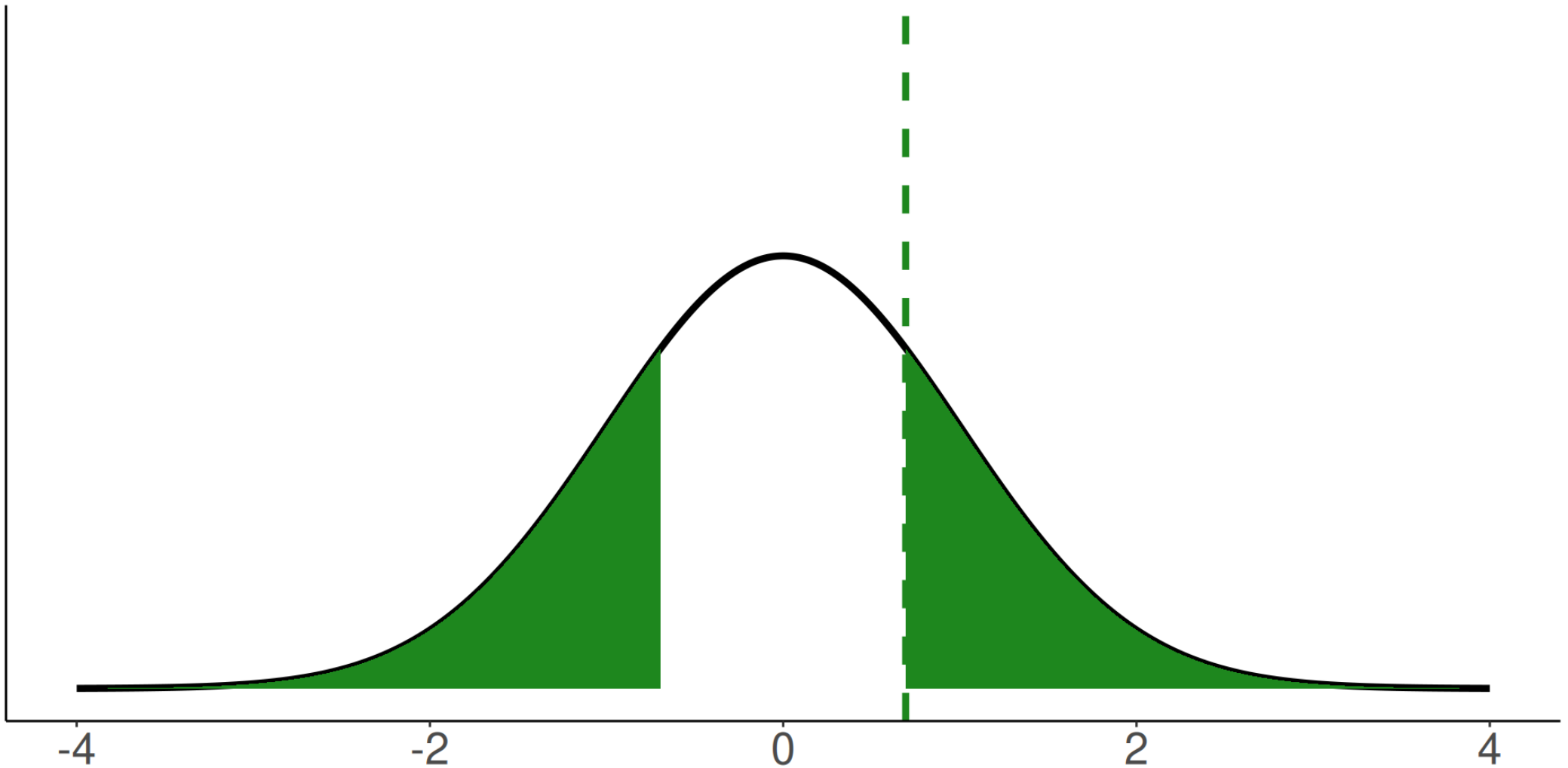
$$z_{obs} = \frac{0.52 - 0.5}{\sqrt{\frac{0.5(1-0.5)}{300}}} = 0.693$$

which is a z-score. How does this compare to the standard normal null distribution?

The test statistic is a z-score, which we can compare to a normal null distribution:



Two-sided p-value:



We get a p-value of 0.49, slightly smaller than the binomial test but still non-significant.

This works because:

- BINS assumptions
- $np(1-p) \geq 10$ for $n = 300$ and $p = 0.5$

Exercises

A survey of 15 high-school students showed that 11 of them use social media. Consider using this data to test hypotheses

$$H_0 : p = 0.8 \quad \text{versus} \quad H_A : p \neq 0.8$$

where p is the true proportion of high-school students who use social media. Use $\alpha = 0.01$.

- Complete the test with a binomial null distribution as well as with a Z test.
- How big would n need to be before we would feel comfortable testing these hypotheses with a Z test?

Solutions/Hints

- The null binomial distribution is $Binom(15, 0.8)$.
- The Z test is not a good test to use here, since $Binom(15, 0.8)$ is not well approximated by a normal. We have $15(0.8)(0.2) < 10$.
- If we want $n(0.8)(0.2) \geq 10$, we would want n to be at least 63.

$$H_0 : p = p_0 \quad \text{versus} \quad H_A : p \neq p_0$$

If we make the BINS assumptions and have $np_0(1-p_0) \geq 10$, we can use a Z-test statistic of

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

and a $N(0,1)$ null distribution.

We can also create a confidence interval with our model, without resampling.

An interval gives us a range of guesses (in this case, different guesses for p).



If we don’t have 10000 resamples, where does the interval range come from?

The standard error of $\hat{p}$:

$$\sqrt{\frac{p(1-p)}{n}}, \quad \text{estimated as} \quad \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

This represents the estimation error when we use $\hat{p}$ to guess p.

Idea:

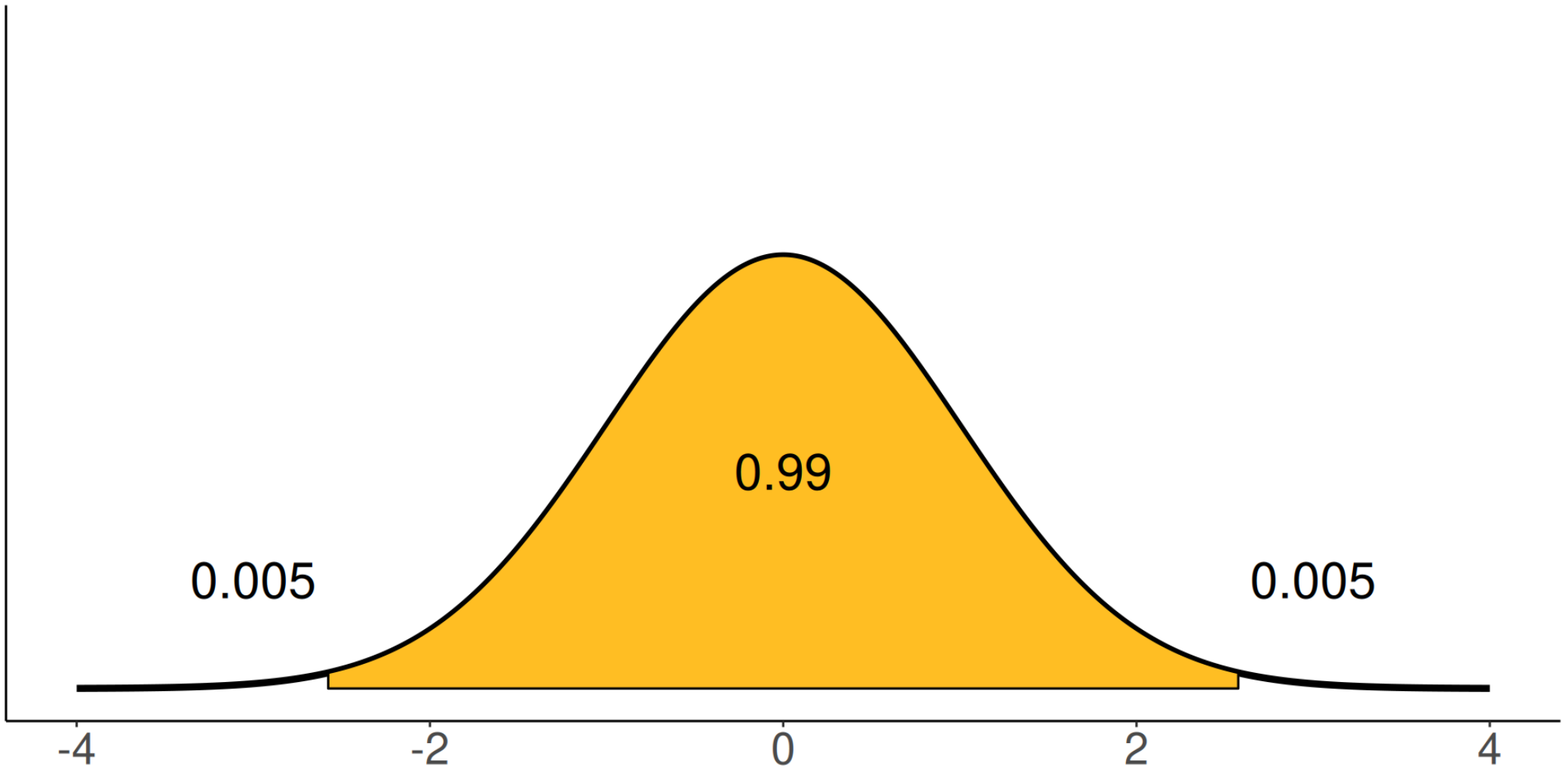
- Center our interval at $\hat{p}$
- Add a margin above and below based on our se

$$\text{Proportion CI} = (\hat{p} - \text{CV} \times \text{SE},\ \hat{p} + \text{CV} \times \text{SE})$$

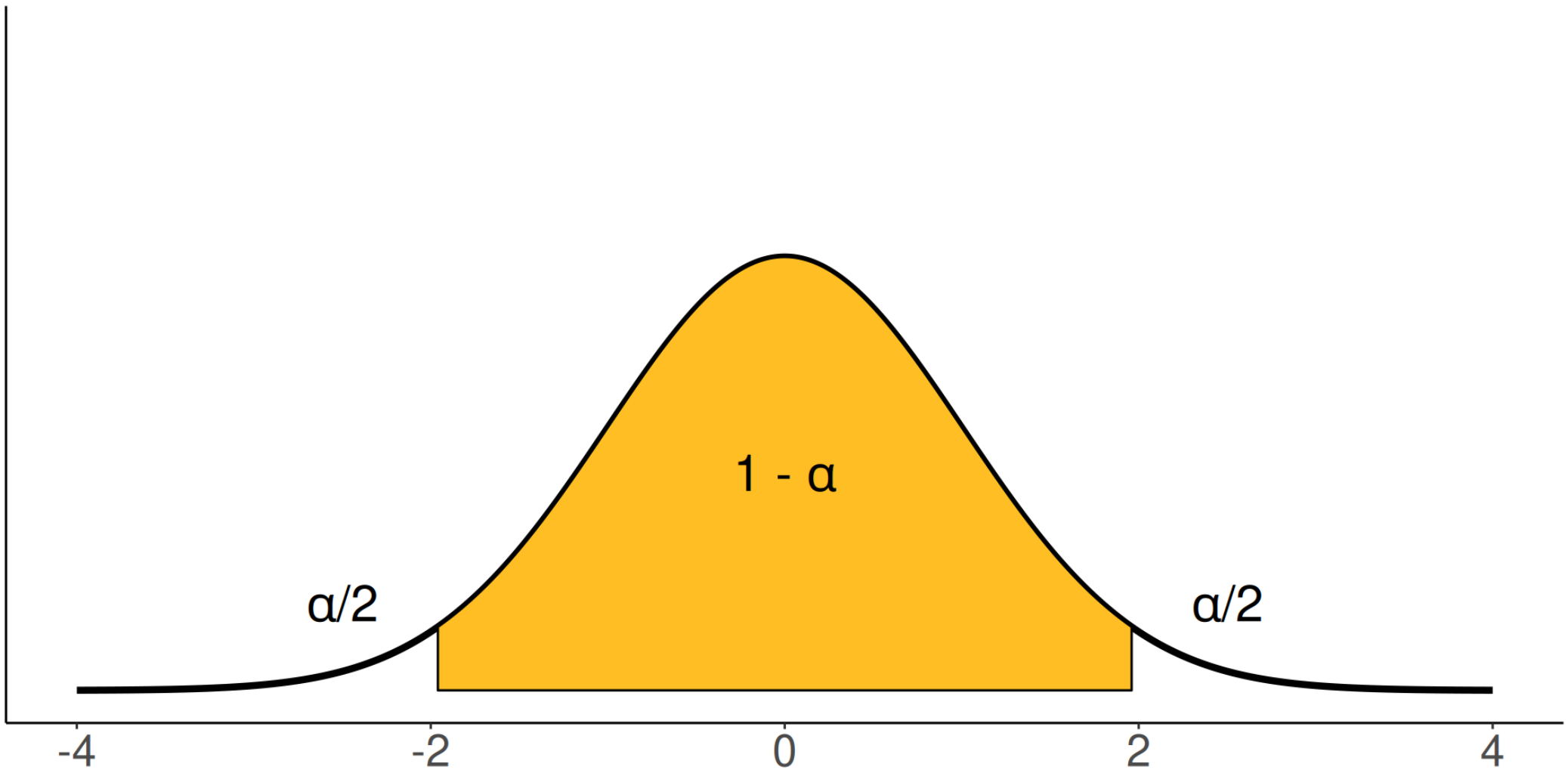
The multiplier, or **critical value**, comes from the confidence level $1 - \alpha$.

“How many error terms do I need to be 99% confident?”

Critical values are percentiles on the normal curve. For a 99% CI, use the 0.5th and 99.5th percentiles.



In general, use the $\alpha/2$ percentiles.



The critical value for 99% confidence is 2.576. In the powerplant example, the estimated standard error is

$$\sqrt{\frac{0.52(0.48)}{300}} = 0.0288$$

We get a 99% confidence interval

$$\Big(0.52 - 2.576(0.0288),\ \ 0.52 + 2.576(0.0288)\Big)$$

$$= (0.446, 0.594)$$

We are 99% confident the true p is within $(0.446, 0.594)$.

In general, CIs made with symmetric probability models (not resampling) are:

$$\text{point estimate} \pm \text{critical value} \times \text{standard error}$$

The specific formulas depend on what method we're using and what our parameter of interest is.

One-sample proportion CI:

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Again, this works if we have:

- BINS assumptions
- $np(1-p) \geq 10$

Exercises

Re-create the powerplant proportion CI, but use 95% instead of 99%.

- What is the normal critical value for 95% confidence?
- Is the resulting interval wider or narrower than the 99% CI?
- What is the difference in the standard error term used in a proportion Z test versus a proportion Z interval? Where does this difference come from?



Solutions/Hints

- The 95% CI uses a critical value of about 1.96, and results in a narrower interval than the one with the 99% CI.
- If we want $n(0.8)(0.2) \geq 10$, we would want n to be at least 63.

Looking ahead

In chapter 11, we'll learn about how to study a difference in two proportions. If you want a head start, consider two sample proportions from two different groups, $\hat{p}_1$ and $\hat{p}_2$.

What is the expected value of $\hat{p}_1 - \hat{p}_2$? Is this quantity normal?